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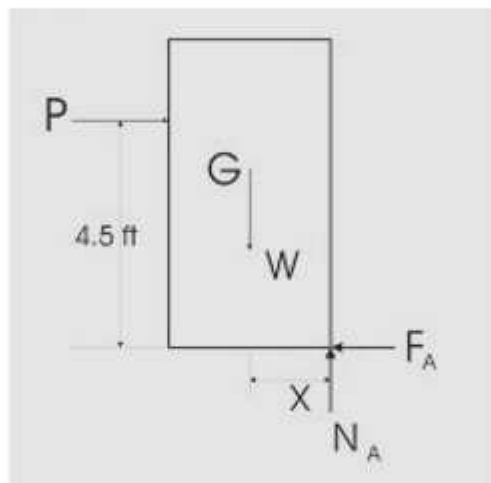
Answer

Given data:

Weight of the crate, $W = 250 \text{ lb}$

Coefficient of friction between crate and the surface, $\mu_s = 0.4$

Draw the free body diagram, of the crate.



Here, F_A is the friction force and N_A is the normal force.

Apply the equilibrium equation for the crate.

$$\sum F_x = 0$$

$$P - F_A = 0$$

$$P - \mu_s N_A = 0 \quad (\text{friction force } F_A = \mu_s N_A)$$

$$P = 0.4 N_A \quad \dots\dots (1)$$

$$\sum F_y = 0$$

$$-W + N_A = 0$$

$$N_A = W$$

$$= 250 \text{ lb}$$

Substitute the values in equation (1).

$$P = 0.4 \times 250$$

$$= 100 \text{ lb}$$

Take moment about point G.

$$\sum M_G = 0$$

$$P(4.5 - 3.5) - (N_A \times x) + (F_A \times 3.5) = 0$$

$$100 \times (4.5 - 3.5) - (250 \times x) + (0.4 \times 250 \times 3.5) = 0$$

$$x = 1.8 \text{ ft}$$

Since, the value of x is greater than 1.5 m. So assumption made is incorrect.

Take the moment about point A.

$$-(P \times 4.5) + (W \times 1.5) = 0$$

$$-(P \times 4.5) + (250 \times 1.5) = 0$$

$$P = 83.33 \text{ lb}$$

Hence, the minimum value of force P is 83.33 lb.

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